School: ..... Campus: .....

Academic Year: ..... Subject Name: ..... Subject Code: .....

Semester: ..... Program: ..... Branch: ..... Specialization: .....

Date: .....

Applied and Action Learning

(Learning by Doing and Discovery)

Name of the Experiement : Architecture Deep Dive – Explore Blockchain Types & Consensus

Objective/Aim:

To study and understand different types of blockchain architectures and the various consensus mechanisms used to validate transactions in a blockchain network.

Apparatus/Software Used:

- Computer/Laptop
- Internet connection
- Blockchain simulators or platforms (e.g., Ethereum testnet, Hyperledger tools)
- Web browser

Theory/Concept:

1. **Blockchain Architecture:**

   Blockchain is a decentralized digital system where data is stored in blocks and connected in a chain using cryptography. Each block contains transaction details, a timestamp, and a link to the previous block, making the data secure and impossible to change. This ensures transparency, trust, and data integrity without needing a central authority.

2. **Types & Consensus Mechanisms:**

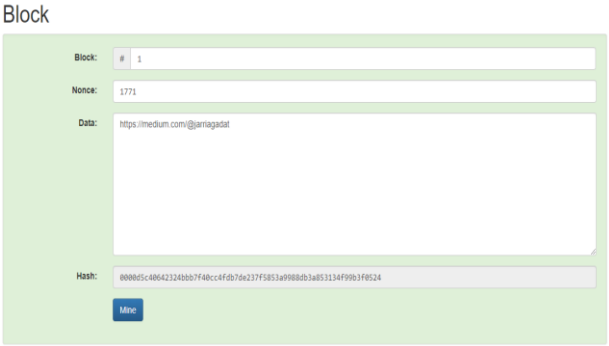
   Blockchain can be of different types such as public (open to all), private (restricted access), and hybrid (combination of both). To verify transactions, blockchain uses consensus mechanisms like Proof of Work (PoW), where computers solve complex problems, and Proof of Stake (PoS), where validators are chosen based on their stake. These methods help maintain security, accuracy, and agreement across the network.

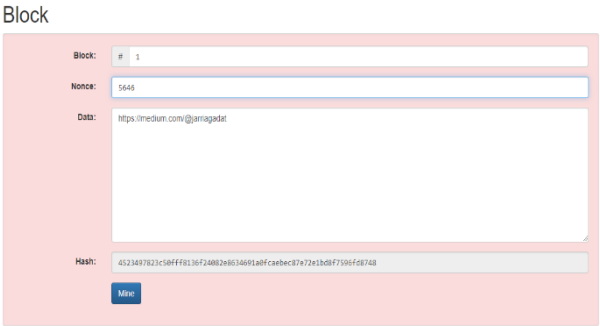
Procedure:

1. Study the basic structure of blockchain architecture.
2. Identify different types of blockchains and their characteristics.
3. Analyze real-world examples of each blockchain type.
4. Explore various consensus algorithms used in blockchain.
5. Compare PoW, PoS, and other mechanisms based on:

- Security
- Speed
- Energy consumption

6. Use online tools or simulators to observe how transactions are validated.





Observation Table:

Blockchain Type	Access Level	Decentralization	Example Use Case
Public	Open	High	Cryptocurrency
Private	Restricted	Low	Enterprise use
Consortium	Limited	Medium	Banking systems
Hybrid	Mixed	Flexible	Government apps

ASSESMENT

Rubrics	Full Mark	Marks Obtained	Remarks
Concept	10		
Planning and Execution/ Practical Simulation/ Programming	10		
Result and Interpretation	10		
Record of Applied and Action Learning	10		
Viva	10		
Total	**50**		